

RESEARCH

Open Access



International Chinese language learners' cultural vocabulary proficiency: a novel scale development and influencing factors analysis

Lijiao Yang^{1*†}, Yifan Lin^{1†}, Panpan Zhou¹, Danyang Qiu² and Luyao Chen^{3*†}

[†]Lijiao Yang and Yifan Lin are co-first authors.

[†]Luyao Chen and Lijiao Yang contributed equally.

*Correspondence:
yanglijiao@bnu.edu.cn;
luyaochen@cbs.mpg.de

¹ School of International Chinese Language Education, Beijing Normal University, Beijing 100875, China

² Beijing National Day School, Longyue, Beijing 100089, China

³ Hong Kong Polytechnic University, Department of Language Science and Technology, Faculty of Humanities, 8 Hung Lok Road, Hung Hom, Kowloon, Hong Kong, China

Abstract

Cultural instruction plays a vital role in international Chinese language education; however, the systematic integration of cultural content into curricula and teaching materials remains underexplored at the empirical level. This study addresses this gap by investigating learners' cognitive understanding of Chinese cultural vocabulary, aiming to support more targeted cultural instruction and curriculum development across diverse learner populations. Employing a corpus-based methodology, the study constructed a specialized corpus of 2329 culturally significant words aligned with international Chinese education standards. From this corpus, 93 representative words were selected across elementary, intermediate, and advanced levels to develop a Likert scale-based assessment tool. The tool's psychometric properties were validated through empirical testing with 255 learners from various cultural backgrounds. Results indicate that learners' Chinese language proficiency, duration of study, and cultural background significantly influence their mastery of cultural vocabulary. The study offers a novel and validated instrument for measuring cultural vocabulary knowledge, laying the groundwork for data-driven cultural syllabus design and instructional strategies. These findings contribute to enhancing intercultural competence among Chinese language learners and provide a scalable framework for advancing cultural instruction in international Chinese language education.

Keywords: International Chinese Language Education, Cultural vocabulary scale, Corpus, Cross-cultural awareness

Introduction

Basic and commonly used words that describe everyday items and activities constitute the foundation of verbal communication. In contrast, cultural words carry profound cultural significance and context, providing unique insights into a culture and its background, values, beliefs, traditions, customs, and history (Nation, 2001; Lantolf & Pavlenko, 1995; Mackey, 2004). For example, the cultural term Dao (道) encapsulates a fundamental concept in Chinese philosophy. It refers to the underlying principle that governs the operation of the universe, often expressed through the dynamic balance of Yin and Yang. When Dao was first introduced to the West, it was commonly translated as "the way." Although this translation captures part of the original meaning, it can

oversimplify or obscure the broader philosophical connotations of the term in its native context. In recent years, it has been translated as Tao or Dao, reflecting its pronunciation, primarily owing to the inherent challenge in precisely defining cultural concepts such as “Dao”, as there is no exact equivalent in English. In second language (L2) learning, cultural vocabulary and everyday common vocabulary are integral components, together forming the primary content of vocabulary acquisition (Laufer & Ravenhorst-Kalovski, 2010; Schmitt, 2008).

It is widely recognized that cultural vocabulary significantly influences reading comprehension, particularly in second language contexts where background knowledge plays a crucial role in meaning construction (Carrell, 1987; Koda, 2005; Benattabou, 2020). In addition, literary works, historical records, and humanities texts often contain vocabulary and expressions tied to specific cultures. Learners’ proficiency in cultural vocabulary in the target language may closely correlate with their overall vocabulary proficiency and reflect their cross-cultural awareness (Citron, 1995; Tišma & Krsmanović, 2018). If second language learners lack familiarity with the meanings and cultural backgrounds of these vocabulary items, they may struggle to fully comprehend the intended meaning and emotions conveyed in the text. Therefore, acquiring a broader range of cultural vocabulary efficiently enables readers to more accurately capture cultural nuances and implications, facilitating a deeper understanding and interpretation of the text (Steffensen et al., 1979; Koda, 1994).

At the same time, cultural vocabulary contains unique cultural connotations and is an important part of cultural teaching. As an important part of international Chinese language education, teaching culture plays an important role in promoting cultural diversity and fostering learners’ global perspectives. Through the teaching of cultural vocabulary, learners not only acquire the meanings of words but also gain deeper insights into the values and modes of thinking embedded in the Chinese language, thereby enhancing their cross-cultural communicative competence. Research on cultural words facilitates the identification of high-frequency, culturally loaded vocabulary, enabling the development of systematic instructional content and promoting the cultural orientation of both textbooks and assessments. By adopting contextualized, hierarchical, and categorized approaches, cultural word instruction can enrich the depth of cultural cognition and pragmatic competence in language learning. In sum, the integration of language and culture has become a widely recognized and core principle in international Chinese language education (Yang, 2019).

In the field of international Chinese language education, vocabulary acquisition has always been a crucial aspect of language and culture teaching. However, most studies on second language vocabulary acquisition have only explored basic vocabulary, with less attention paid to measuring L2 learners’ mastery of cultural vocabulary (Laufer, 1998; Crossley et al., 2011). Exploring L2 learners’ knowledge of cultural vocabulary is equally vital in designing teaching content, activities, and strategies. Nevertheless, the specific cultural points and cultural vocabulary that should be integrated into teaching materials are still widely debated (Wu, 2022). In 2022, the Chinese Culture and Society Reference Framework in Chinese Language International Education was launched. It is the first programmatic document on culture teaching in international Chinese language education and demonstrated the new concepts and ideas of culture teaching

in international Chinese language education by defining the scope of culture teaching, describing the goals and contents of culture teaching, sorting out cultural items, exemplifying cultural teaching points, and dividing the teaching levels (Zu, 2023). The proposal of the framework provided a standardized operational guideline for culture teaching and provided new reference bases for the writing of culture teaching materials and for the study of cultural vocabulary.

It is worth considering the measurement of learners' understanding of cultural words across different countries, cultural backgrounds, and levels of Chinese L2 proficiency. Such assessment would help reveal how well learners at varying stages have acquired culturally embedded language elements. In this case, such a measurement tool will not only help develop cultural syllabi with empirical evidence but also be beneficial for promoting the relevance and scientificity of incorporating cultural elements into teaching practices. Therefore, this study aimed to fill this gap by developing a Chinese cultural vocabulary scale for learners of Chinese as a foreign language.

Literature review

Definition of cultural vocabulary

With the surge in cross-cultural communication studies during the late 1960s, the concept of "cultural vocabulary" proliferated alongside the "cultural fever" in language teaching research. During this period, cultural vocabulary, also referred to as culture-loaded vocabulary or the cultural gap vocabulary, gained prominence in the field. American linguist and translator Nida (1964) was among the earliest scholars to define and classify cultural vocabulary in his monograph, providing a systematic overview of the cultural factors integrated into the linguistic structure, including ecological, linguistic, religious, material, and social culture.

Subsequently the discourse on the definition of cultural words has grown and can be divided into two categories depending on the field of study: (a) the domain of second language acquisition and translation; (b) the socio-cultural domain. In the fields of second language acquisition and translation, scholars consider entries with similar primary meanings in both L1 and L2 but differing connotations as cultural vocabularies (Lado, 1972; Dagut, 1978; Baker, 2018). Dagut, for example, restricted cultural vocabulary to nouns and categorized them into environmental, cultural, lexical, and syntactical voids. Mona took a more restrictive stance, that only words lacking equivalents in the target language can be deemed cultural vocabulary. In contrast, in the sociocultural domain, cultural vocabulary is defined as those words originating from a specific sociocultural context, reflecting the cultural and spiritual essence of the language community (Newmark, 1998; Hartmann & James, 2002).

As the concept of "cultural vocabulary" permeated Chinese academia, three distinct viewpoints on its definition emerged: cultural meaning, cultural additional meaning, and cultural content theories (Zhao, 2009). According to cultural meaning theory, the cultural significance beyond a word's literal meaning constitutes its cultural meaning, and words embodying such significance are labeled cultural vocabularies, exemplified by terms such as "QueQiao (in Chinese 鹊桥)" and "BaoGong (in Chinese 包公)" (Chang, 1995; Meng, 1992). Conversely, the theory of additional cultural meaning asserts that only words encompassing both conceptual and cultural meanings qualify as cultural

vocabulary, encompassing specific national cultural information while referencing objects (Zhang, 2003). Additionally, according to cultural content theory, only words originating from a specific cultural context and reflecting unique national cultural content are considered cultural vocabularies, such as “TaiJiQua” and “Zhu Geliang” (Wang, 1991). The prevailing academic view favors the cultural content theory, defining Chinese cultural vocabulary as words that reflect the distinctive cultural context of the Chinese nation.

The concept of cultural words has sparked discussions from various academic perspectives, resulting in significant debates and differences in their definitions (Risager, 2006). For the sake of precision and quantifiability, this study considers Chinese cultural words as linguistic expressions within the Chinese language that encapsulate profound social and cultural implications and are intimately intertwined with the psyche, traditions, and societal transformations of the Chinese people. They epitomize the distinctive essence and enduring legacy of Chinese culture. These words process precise connotations, evocative connotations, figurative import, and symbolic significance, constituting a dynamic cornerstone of the lexicon.

Cultural vocabulary and second language acquisition

Research on cultural vocabulary in second language acquisition has been highly active and has produced significant results. In developing cultural vocabulary lists, researchers predominantly rely on subjective selection methods. However, some scholars have constructed graded cultural vocabulary lists by creating cultural corpora and integrating word frequency data with expert judgment (Tüfekçioğlu, 2021). To measure cultural vocabulary, existing studies often utilize questionnaire surveys or employ cultural vocabulary scales constructed on the basis of situational semantic judgment during participants' testing (Altakhaineh & Zibin, 2014; Benattabou, 2020). However, experimental outcomes tend to rely heavily on the subjective judgment of test subjects and lack objectivity. With respect to the teaching application of cultural vocabulary, almost all research indicates a bidirectional promotional effect between the acquisition of cultural vocabulary and second language learning (Byon, 2007).

Research on Chinese cultural vocabulary among CS/FL learners

Since the late 1990s, research on Chinese cultural vocabulary among CS/FL learners has gradually increased, focusing on the ontology of cultural vocabulary and translation strategies and methods (Zhao, 2010; Dang, 2015; Yu, 2016). Over the past decade, cultural vocabulary teaching has gained widespread attention in international Chinese education, although research outcomes in this area remain limited. Existing studies primarily employ qualitative methods to assess learners' acquisition of cultural vocabulary and offer recommendations for organizing cultural vocabulary in textbooks and implementing teaching strategies.

Zhang (2012) surveyed 83 international students at intermediate and advanced levels of Chinese language proficiency via a questionnaire comprising 15 judgments, 25 multiple-choice, and 10 sentence construction questions, supplemented by interviews to assess cultural word awareness across different nationalities. Li (2000) devised the culture word test to examine the learning patterns of 23 Korean students regarding cultural

words, offering three to six answer choices under each word to gauge students' comprehension. While these experiments represent initial efforts to measure the acquisition of cultural words, they exhibit limitations in depth and breadth.

Research on lexical measurement

Lexical competence involves an individual's knowledge and ability to use words in a language, categorized into dimensions such as the breadth of vocabulary and depth of vocabulary dimensions. The breadth of vocabulary dimension concerns the number of words a person comprehends in a language, typically assessed through vocabulary size tests. Similarly, the size of a student's vocabulary, indicating the breadth of their lexical knowledge, serves as a commonly used metric for evaluating their language proficiency (Schmitt et al., 2001; Nation, 2005). Moreover, investigating students' vocabulary size assessment offers valuable insights into their lexical competence, particularly their breadth of vocabulary. Such research is crucial for textbook development, teaching evaluation, course standardization, and acquisition studies.

In recent years, scholarly attention has focused on vocabulary testing research and its pedagogical applications in international Chinese education (Zhao, 2003; Jiang & Liu, 2004; Hao & Fan, 2008; Zhang et al., 2021). Tests typically select test words from sources such as the "Chinese Proficiency Vocabulary and Character Level Outline" and the "Modern Chinese Frequency Dictionary" (Qian, 2002; Zhang, 2006; Shen, 2009) to assess learners' breadth of vocabulary mastery at different difficulty levels, with selection criteria being more subjective. In addition, in the field of second language teaching, previous vocabulary measurement methods have often relied on translations or multiple-choice questions. However, cultural vocabulary, a distinct category with more ambiguous word boundaries, reflects students' comprehension and application of cultural concepts. Conversely, as a tool for gauging subjects' attitudes and subjective experiences, the Likert scale is better suited to handle ambiguity and provide richer feedback. Therefore, a Likert scale might serve as a more effective method for scientifically assessing the size of cultural vocabulary.

Addressing gaps in cultural vocabulary assessment: tools, corpora, and learner factors

Despite the growing interest in and recognition of the importance of cultural vocabulary in second language acquisition, there remains a significant gap in the field regarding a standardized measure for assessing cultural vocabulary knowledge, especially in Chinese language education. Existing studies have focused predominantly on assessing everyday vocabulary, leaving a substantial void in the evaluation of cultural words in Chinese as a second language. This includes both the lack of suitable assessment tools and a comprehensive, corpus-based cultural vocabulary database. In this context, two potential research questions for future investigations are as follows:

RQ1. How to design and implement standardized assessment tools for cultural vocabulary that more accurately and swiftly measure the extent of learners' mastery of cultural words?

RQ1 is actually twofold: On the one hand, it focuses on what the assessment tools for cultural vocabulary might look like, and this was answered by "Construction of

instruments” section and “Target word test: questionnaire design and collection” section. In these two sections, the cultural vocabulary database was constructed and then the target word test was established. On the other hand, the validity of such a novel tool was systematically evaluated as reported in “Results: analysis of cultural vocabulary tests and influencing factors” section and “Validation using the IRT model: content aspect of construct validity” section.

RQ2. What factors would influence learners’ mastery of cultural words? This research aimed to uncover the diverse influences on learners’ acquisition of cultural vocabulary, such as their rank of Chinese language proficiency (HSK), the impact of popular culture, and the effects of their national backgrounds.

Answering RQ1 simultaneously necessitates answering RQ2. In particular, RQ2, focusing on the potential factors influencing learners’ mastery of the cultural vocabulary, was simultaneously assessed as a key compositional part of the validity tests: The difficulty level of the cultural vocabulary was tested in “Empirical validity: difficulty level effect of cultural vocabulary” section, and the effects of Chinese character circles and Chinese proficiency were tested in “Empirical validity: effects of Chinese character circle and Chinese proficiency” section. The content aspect of construct validity was systematically tested in “Validation using the IRT model: content aspect of construct validity” section.

Methods

Participants

We collected data from 255 Chinese language learners residing domestically and internationally, representing 26 countries. Following data processing, 191 valid responses were retained, whereas 64 valid responses were excluded, mainly because of consistent or nonsensical responses and low Chinese proficiency (HSK level 3 or below). The test in this project presented questions in Chinese characters without pinyin annotation, making it challenging for learners with an HSK level of 3 or below, leading to their exclusion from the analysis. Among the remaining 191 subjects, 29 were male and 162 were female. They included 111 individuals with Chinese character-based language backgrounds from countries such as Japan, South Korea, and Vietnam, and 80 with non-Chinese character-based language backgrounds from Thailand, Russia, Mongolia, and 23 others. Additionally, 125 participants belonged to the intermediate-level group (HSK level 4), whereas 66 were in the advanced-level group (HSK levels 5 and 6). Table 1 presents the background information and self-assessment scores of the participants.

Construction of instruments

Cultural vocabulary database construction

Owing to the absence of a Chinese cultural word frequency List and related cultural word outline, we utilize the dynamic corpus of Chinese international teaching materials as our data source. Compiled by Yang Iijiao Research Group at Beijing Normal University, the corpus includes 5413 texts within the realm of international Chinese language education, covering topics ranging from daily life to medicine, business, science, and technology, as well as traditional culture, ensuring broader coverage within the field of international Chinese language education. Additionally, we supplement the lexical grade

Table 1 Background information and self-assessment scores of research participants

Native language writing background	Chinese proficiency	Male	Female	Learning duration (years)	Primary mean	Intermediate mean	Advanced mean	Overall mean
Chinese character background	Intermediate	6	64	5 (1.17)	4.07 (0.58)	3.69 (0.66)	3.40 (0.65)	3.99 (0.63)
	Advanced	5	36	5.76 (0.60)	4.38 (0.52)	4.11 (0.67)	3.51 (0.72)	4.07 (0.64)
Non-Chinese character background	Intermediate	12	43	4.16 (0.83)	3.91 (0.57)	3.55 (0.69)	3.00 (0.66)	3.58 (0.63)
	Advanced	6	19	4.93 (0.06)	4.32 (0.58)	4.03 (0.71)	3.40 (0.74)	3.99 (0.69)

and cultural attributes of selected words by referencing the “International Chinese Education Chinese Proficiency Level Standards (2021) (hereinafter referred to as the ‘Outline’)” and the “Concise Dictionary of Traditional Chinese Culture (2021).”

The “Outline,” the State Language Commission’s main comprehensive description of the criteria for assessing foreign learners’ Chinese language skills and proficiency, contains 11,092 vocabulary words and is divided into beginner, intermediate, and advanced levels on the basis of their difficulty levels. To align with the objectives, this research selects words falling outside the scope of grades 1 to 9 outlined in the vocabulary, since these words typically encompass more cultural information.

Additionally, entries in the “Concise Dictionary of Traditional Chinese Culture” are derived from words and phrases reflecting exceptional traditional Chinese culture, incorporating relevant basic concepts, knowledge words and phrases, and figures and works. It includes approximately 3600 words from Confucianism, Taoism, Buddhism, and other schools of thought. Therefore, the “Concise Dictionary of Traditional Chinese Culture” and expert knowledge were extensively used to select cultural vocabulary in this research.

Corpus processing involved four main stages: (1) scanning and optical character recognition (OCR); (2) cleaning and standardization of digital texts. This study employs jieba word segmentation, word frequency statistics technology in Python, and manual proofreading to identify out-of-scope cultural words and their frequencies in Chinese international textbooks; (3) metadata annotation including title, source, genre, and theme; and (4) lexical processing through segmentation and part-of-speech tagging. This process constructs the “Out-of-Scope cultural vocabulary Frequency List for International Chinese Textbooks,” comprising a total of 2329 words.

Word sampling

In line with the measurement objectives of this study, we comprehensively considered factors such as word frequency level, complexity of Chinese characters, lexical semantic category, and degree of cultural communication (including regional breadth and commonality in daily life) in the “Out-of-Scope cultural vocabulary frequency list for international Chinese textbooks.” Coupled with expert knowledge, we manually selected 93 target words. To assess the test’s completeness and enhance scale effectiveness, we manually extracted 13 common noncultural words from each level of the outline as

interference items, referring to information on word frequency and vocabulary level. The specific considerations for the cultural vocabulary test are as follows:

Word frequency: To further investigate the impact of word frequency on the comprehension of cultural words, this research calculates the occurrence frequency of cultural words in the textbook corpus by using a Python script. Notably, the frequency of cultural words based on corpus statistics can partly reflect their actual usage in daily communication.

Word complexity: For cultural vocabulary within the same semantic category, the complexity of Chinese characters, which is determined by the number of strokes, also influences learners' cognition. For example, "couplet" (in Chinese: 对联) and "spring couplet" (in Chinese: 春联) both denote a type of red festive element "nianhong" commonly pasted during the Spring Festival. However, the former presents significantly greater character complexity than the latter.

Degree of cultural transmission: The degree of cultural transmission includes the breadth of regional dissemination and commonality in daily life, indicating the public's awareness and understanding of various cultural vocabularies. For example, owing to historical exchanges with neighboring countries, traditional Japanese and Korean cultures exhibit strong Confucian influences, with Confucianism and etiquette culture persisting in everyday life. In contrast, within the Chinese character cultural circle, traditional Chinese cultural vocabularies such as "yin and yang" and "five elements" enjoy greater recognition than non-Chinese character cultural regions.

To ensure the scientific rigor and validity of the cultural vocabulary list, this study utilizes the "Big Data Report on the Popularity of Traditional Chinese Culture Transmission (2022)" [Big Data Report on the Spread of Chinese Traditional Culture (2022), Phoenix, 27 February 2023. <https://news.ifeng.com/c/8NkTy7jHLuI>] as a reference, categorizing words into high- and low-transmission groups. As the first comprehensive study of the transmission ecology of traditional Chinese culture in China, this report employs big data tools such as Phoenix Network questionnaire surveys, the Baidu Index, Sina Public Opinion, and the WeChat Index to analyze the prevalence of traditional Chinese cultural content across the entire internet. It also investigates over 400 traditional cultural keywords encompassing cultural commonalities, classic works, life skills, customs, moral values, and philosophical concepts. Notably, the words "calligraphy," "Three Kingdoms," "health preservation," "Water Margin," "Confucianism," "Historical Records," "Cao Cao," and "Sun Wukong" are among the top ten keywords in information, classic reading, and general knowledge figures. These words are included in this word list as high-transmission cultural vocabulary.

Therefore, this report offers insights into the current status and impact of traditional culture transmission in contemporary society from various dimensions and perspectives, providing robust data support and a theoretical foundation for this study. With respect to the factors of word frequency, word complexity, and degree of cultural transmission discussed earlier and considering the expertise of international Chinese language instructors and the Linguistic proficiency of native speakers, cultural words are classified into three categories. These categories include 25 primary cultural words, 40 intermediate cultural words, and 28 advanced cultural words.

Vocabulary semantic category: Drawing from Zhang Dainian's four-level theory of cultural structure presented in "An Introduction to Chinese Culture," this research delineates the semantic categories of cultural vocabulary as follows (as shown in Table 2):

Distractor words: To improve the predictive accuracy of participants' responses, we included 13 distractor words from the international Chinese proficiency level vocabulary outline, organizing them into high, medium, and low tiers.

Target word test: questionnaire design and collection

In the "Chinese and Foreign Cultural Exchange and Dissemination" course, we distributed a preliminary-designed cultural vocabulary scale to 14 students from Japan, South Korea, Ukraine, and Russia. On the basis of the statistical results and student feedback, we refined the original questionnaire, resulting in a cultural word scale comprising 93 cultural words and 13 interfering words. To gauge the depth of subjects' understanding of cultural words, this research employs a Likert scale as the measurement method. Additionally, to assess the seriousness of the participants in answering the questions, we included the extremely low-difficulty distractor "room" (in Chinese: 房间) as a test item. Failure to select "5-very familiar" for this test item indicates a lack of seriousness in answering.

In the Sojump platform (www.wjx.cn; also known as Wen Juan Xing (问卷星) in Chinese), an online tool for questionnaire distribution and data collection, we distributed questionnaires to 303 Chinese as a second language learners across various countries, including Japan, South Korea, Thailand, Laos, Indonesia, the Philippines, Belarus, Egypt, Mongolia, Singapore, and Myanmar. Additionally, we employed the same tool to distribute 84 separate questionnaires to Vietnamese learners of Chinese as a second language. In total, we collected 191 valid data points from both methods.

It is also noteworthy that the cultural vocabulary list developed in this paper mainly serves as a test to measure the respondents' understanding and mastery of cultural words. Based on the results of the test, it allows teachers to have an approximate understanding of the students' level of cultural awareness, and based on which subsequent targeted instruction can be developed.

Results: analysis of cultural vocabulary tests and influencing factors

All data were normally distributed (tested by the Shapiro–Wilk tests, $p > .05$) and, therefore, did not violate the statistical assumptions of the subsequent parametric statistic tests (such as t test and analyses of variance, ANOVA).

Empirical validity: difficulty level effect of cultural vocabulary

Table 3 shows the mean self-assessment scores of participants on cultural vocabulary at different levels. Learners' mean self-assessment scores decrease progressively

Table 2 Semantic categories of cultural vocabulary

The first cognitive layer	The internet and the present era, architecture, historical culture, and cultural arts
The second cognitive layer	Cultural views, cultural products, celebrities, behaviors, and customs, as well as mascots
The third cognitive layer	Administration and system, language and script, titles and communication

Table 3 Mean self-assessment scores for primary, intermediate, and advanced cultural vocabulary

	Primary	Intermediate	Advanced
Mean	4.12	3.79	3.24
Median	4.20	3.81	3.22
Standard deviation	0.59	0.71	0.70
Minimum value	2.63	2.10	1.56
Maximum value	5	5	4.63

from primary to intermediate and advanced levels. Similarly, repeated measures analysis of variance revealed a significant main effect of difficulty level on the mean self-assessment score, $F(2,380) = 573$, $p < 0.001$, $\eta^2p = 0.75$. Post hoc multiple comparisons revealed significant differences in self-assessment means between primary, intermediate, and advanced cultural vocabularies: primary vs. intermediate, $t(190) = 15.4$, $p < 0.001$, Cohen's $d = 1.11$; primary vs. advanced, $t(190) = 28.8$, $p < 0.001$, Cohen's $d = 2.08$; intermediate vs. advanced, $t(190) = 21.3$, $p < 0.001$, Cohen's $d = 1.54$.

Empirical validity: effects of Chinese character circle and Chinese proficiency

To determine the influence of the Chinese character circle and Chinese proficiency on cultural vocabulary self-assessment scores, we conducted four regression analyses. The independent variables included gender, learning duration, Chinese character circle background, and Chinese proficiency, while dependent variables were learners' self-assessment scores on primary, intermediate, advanced, and overall cultural vocabulary. In each model, there was no significant multicollinearity among the independent variables. Congruently, the regression analysis results revealed that models incorporating the aforementioned four independent variables significantly accounted for the variance in self-assessment scores for cultural vocabulary at all levels: primary, $F(4,186) = 6.78$, $p < 0.001$, $\eta^2p = 0.13$, $R^2 = 0.10$, adjusted $R^2 = 0.08$; intermediate, $F(4,186) = 8.16$, $p < 0.001$, $\eta^2p = 0.15$, $R^2 = 0.15$, adjusted $R^2 = 0.13$; advanced, $F(4,186) = 5.17$, $p < 0.001$, $\eta^2p = 0.10$, $R^2 = 0.13$, adjusted $R^2 = 0.11$; overall, $F(4,186) = 6.41$, $p < 0.001$, $\eta^2p = 0.12$, $R^2 = 0.12$, adjusted $R^2 = 0.10$.

Table 4 illustrates the significant predictive effects of Chinese proficiency, Chinese character circle, and Chinese learning duration on self-assessment scores for primary, intermediate, advanced, and overall cultural vocabulary. Notably, self-assessment scores for cultural vocabularies increase with longer learning duration and enhanced Chinese proficiency, while learners within the Chinese character circle exhibit higher self-assessment scores compared to those outside it. Besides, we used the relative importance index to assess the significance of each independent variable on the dependent variable, with larger values indicating greater importance. Similarly, Chinese proficiency emerges as the most crucial factor for self-assessment scores across all levels and overall, followed by learning duration, with the impact of the Chinese character circle being slightly weaker. Therefore, gender does not significantly predict cultural vocabulary self-assessment scores.

Table 4 Summary of regression analysis results

Dependent variable	Independent variable	Regression coefficient	Standard error	t	p	Standardized regression coefficient	95% confidence interval	Importance
Primary	Gender	0.12	0.12	1.03	0.30	0.20	−0.19, 0.59	0.07
	Duration	0.09	0.04	2.10	0.04	0.17	0.01, 0.33	0.24
	Chinese proficiency	0.29	0.09	3.12	<0.01	0.49	0.18, 0.79	0.53
	Chinese character circle	0.19	0.09	2.05	0.04	0.32	0.01, 0.63	0.16
Intermediate	Gender	−0.03	0.14	−0.20	0.84	−0.04	−0.42, 0.35	0.01
	Duration	0.17	0.05	3.25	<0.01	0.26	0.1, 0.42	0.40
	Chinese proficiency	0.32	0.11	2.90	<0.01	0.44	0.14, 0.75	0.46
	Chinese character circle	0.27	0.11	2.45	0.02	0.38	0.07, 0.68	0.14
Advanced	Gender	−0.02	0.14	−0.16	0.87	−0.03	−0.43, 0.36	0.01
	Duration	0.12	0.05	2.33	0.02	0.19	0.03, 0.35	0.30
	Chinese proficiency	0.25	0.11	2.27	0.02	0.36	0.05, 0.67	0.44
	Chinese character circle	0.28	0.11	2.49	0.01	0.40	0.08, 0.71	0.26
Overall	Gender	0.00	0.13	0.02	0.98	0.00	−0.39, 0.4	0.01
	Duration	0.13	0.05	2.69	0.01	0.22	0.06, 0.38	0.35
	Chinese proficiency	0.28	0.10	2.70	0.01	0.42	0.11, 0.73	0.48
	Chinese character circle	0.23	0.10	2.22	0.03	0.35	0.04, 0.66	0.16

Validation using the IRT model: content aspect of construct validity

The content aspect of construct validity encompasses content relevance, representativeness, and technical quality. Content relevance measures the alignment of test items with the measured construct, which in this study was the perceived knowledge of vocabulary word form–meaning relationships. As outlined in the methods section, target items were meticulously chosen from a cultural vocabulary dictionary via a stratified sampling approach, ensuring their relevance to the construct domain. Representativeness can be assessed through item strata (formula = $(4G_{\text{item}} + 1)/3$), indicating the number of significantly different levels of item difficulty. The item strata statistic was 10.07, surpassing the threshold level of 2. This suggests that the cultural knowledge scale encompasses more than 10 statistically distinct difficulty levels, bolstering the scale's representativeness.

Furthermore, we evaluated technical quality through item correlations, fit statistics, and local independence. The item correlations revealed positive point–measure correlations across all the items, with values ranging from 0.29 to 0.67. Fit statistics indicated the infit t statistics between 0.69 and 1.61 and the outfit t statistics between 0.56 and 1.89. Local independence was assessed via standardized residual correlations, with most item pairs showing residuals ranging from −0.39 to 0.63. Notably, only one item pair, 13 (in Chinese 老爷) and 65 (in Chinese 网购), exhibited a residual correlation of 0.71, slightly exceeding the threshold of 0.70. Overall, these findings suggest that most items demonstrated satisfactory technical quality.

Substantive aspect of construct validity

Examining the substantive aspect of construct validity involved assessing item hierarchy and person-fit statistics. Items tended to increase in difficulty with higher frequency levels. A one-way ANOVA confirmed the significant main effect of vocabulary frequency level on item difficulty, $F(2, 90) = 15.8, p < .001, \eta^2 p = 0.26$. Post hoc analysis revealed significant differences in item measures between each level: high vs. medium, $t(90) = 3.42, p < 0.01$, Cohen's $d = 0.79$; high vs. low, $t(90) = 6.58, p < 0.001$, Cohen's $d = 1.53$; and medium vs. low, $t(90) = 3.45, p < 0.01$, Cohen's $d = 0.73$. Regarding person-fit statistics, no misfit (outfit/infit $t > 2.0$ or < -2.0) was observed among 188 out of 191 test takers, indicating strong consistency between each person's response pattern and the item hierarchy. Furthermore, the Wright maps illustrated a roughly normal distribution of item difficulties and personal abilities, covering most of the distribution. These findings suggest that the scale demonstrated acceptable substantive construct validity (as shown in Table 5 and Fig. 1).

Structural aspect of construct validity

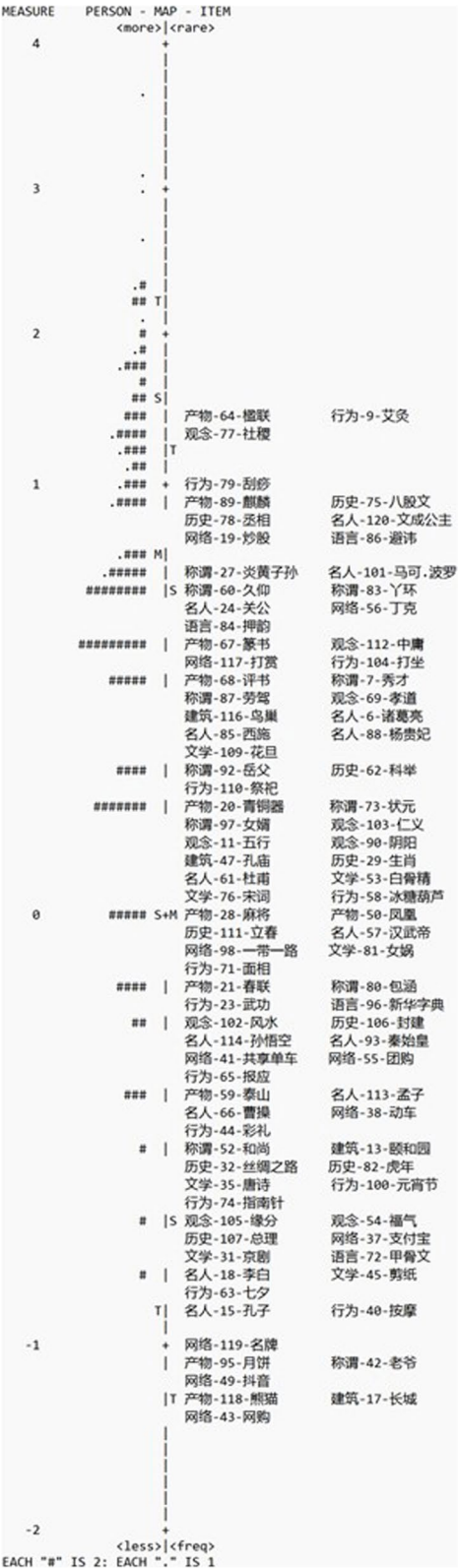
Examining the structural aspect of construct validity involved assessing the unidimensionality of the construct. Linacre's (1995) approach suggests using principal component analysis of standardized residuals. First, the Rasch model explained 44.8% of the variance in the residuals. Second, the factor sensitivity ratio, calculated as the eigenvalue of the first contrast divided by the eigenvalue of the Rasch model, was only 5.5% (Wright & Stone, 2004). Third, the eigenvalues of other contrasts appeared to reach an asymptote at the first contrast. While the eigenvalue of the first contrast exceeded the chance level of approximately 2, a thorough examination of items with loadings greater than .40 and smaller than $-.40$ in the first contrast did not reveal a common meaning (熊猫-panda; 长城-Great Wall; 网络-internet; 月饼-mooncake; 老爷-Lord; 京剧-Peking Opera; 名牌-famous brand; 抖音-Tiktok; 总理-prime minister; 剪纸-paper cut; 甲骨文-inscriptions on bones or tortoise shells of the Shang Dynasty; 丝绸之路-The Silk Road), suggesting that they are unlikely to form a separate dimension. These findings indicate that the scale's unidimensionality was acceptable in both groups.

Generalizability aspect of construct validity

Examining the generalizability aspect of construct validity involves assessing the extent to which the test generalizes across different population groups, such as male and female test-takers. With respect to item calibration invariance, a uniform differential item functioning (DIF) analysis was conducted to determine whether item calibration varied between male and female test-takers in the two groups. A Mantel-Haenszel test identified significant differences between male and female participants in eight items (

Table 5 Summary of the item measures in different frequency levels of vocabulary items

Frequency level	Number	MEAN	SD	SE
High	28	− 0.39	0.47	0.09
Medium	40	0.02	0.55	0.09
Low	25	0.40	0.50	0.10



麒麟-Chinese unicorn; 老爷-Lord; 立春-beginnings of Spring; 文成公主-Princess Wencheng; 关公-Guan Gong-a famous historical martial officer; 网购-online shopping; 艾灸-moxibustion; 甲骨文-inscriptions on bones or tortoise shells of the Shang Dynasty). However, most item calibrations on the cultural knowledge scale did not significantly differ between male and female test-takers. With respect to the reliability of the cultural scale, the overall results met the rule of thumb: Cronbach's alpha reliability = 0.98, person separation = 5.04, person reliability = 0.96, item separation = 7.30, and item reliability = 0.98.

Discussion

The study establishes an international Chinese educational cultural vocabulary corpus and tests it on the "International Chinese Education Chinese Proficiency Level Outline." Experimental verification reveals several key findings: (1) The cultural vocabulary test constructed in this research demonstrates strong performance in terms of reliability, structural validity, and empirical validity, effectively measuring the breadth of knowledge of Chinese cultural vocabulary by Chinese learners. (2) Chinese proficiency, national cultural background (Chinese character circles/non-Chinese character circles), and Chinese learning duration significantly predict the self-assessment scores of primary, intermediate, advanced, and overall cultural vocabularies. (3) Chinese proficiency has the greatest effect on the measurement results of cultural vocabularies, followed by learning duration and native language background (Chinese character circle/non-Chinese character circle), whereas gender has no significant predictive effect.

In contrast to prior studies, this research introduces a novel approach to gauging the breadth of cultural vocabulary among second language learners by utilizing a Likert scale, presenting a new perspective for assessing cultural word proficiency. Past studies have usually used questionnaires or situational semantic judgment tests to measure learners' cultural vocabulary competence (Altakhaineh & Zibin, 2014; Benattabou, 2020). The experimental results rely heavily on subjective judgments and lack objectivity. In addition, the word lists constructed in this paper quantify the differences in effect sizes between tiers, providing a precise empirical basis for cultural vocabulary gradient differentiation. Past word lists such as the one constructed tend to differentiate between tiers based on the subjective judgment of the researcher or word frequency information, which is not objective and scientific (Tüfekçioğlu, 2021).

This study proposes that implementing this cultural vocabulary scale could aid educators and researchers in evaluating learners' cultural vocabulary skills, guiding curriculum design, facilitating teaching assessments, conducting research, and improving learners' vocabulary proficiency and cross-cultural understanding. Developing a cultural vocabulary scale rooted in international Chinese language education is a significant contribution to assessing the cultural vocabulary proficiency of Chinese second language learners.

Interpretation of the Chinese cultural vocabulary scale

The cultural word list designed in this paper has been tested to accurately and efficiently measure learners' mastery of cultural vocabulary.

Cognitive vocabulary hierarchy

Interpreting the Chinese cultural vocabulary scale reveals a cognitive hierarchy structured into 12 categories, which are further divided into three cognitive tiers or "circles" according to learners' cognitive scores, revealing distinctive trends in their comprehension of various cultural words.

Analogously, the primary cognitive tier encompasses high-frequency Chinese vocabulary found in textbooks for foreign learners, alongside data derived from the National Language Commission corpus. This tier predominantly includes lexemes associated with contemporary phenomena and traditional domains such as architecture, historical culture, and cultural arts. Noteworthy terms include Beijing Tang poetry (in Chinese: 唐诗), Chinese zodiac (in Chinese: 生肖), TikTok (in Chinese: 抖音), and Confucius temple (in Chinese: 孔庙). Despite not being part of traditional international Chinese textbooks, these words hold significant relevance to the daily Lives of modern youth, both online and offline, resulting in the highest recognition levels, as evidenced by the average cognitive score of 3.8831.

Congruently, the second cognitive layer encompasses cultural views, cultural products, celebrities, behaviors and customs, and mascots. Words such as Yin and Yang (in Chinese: 阴阳), Spring Festival couplets (in Chinese: 春联), Confucius (in Chinese: 孔子), Wugong (in Chinese: 武功), and panda (in Chinese: 熊猫) are intertwined with historical and cultural narratives, as well as elements of food culture and tourism. While learners recognize these words, their cognitive precision is slightly lower than the first layer, leading to an average cognitive score of 3.6108.

The third and final cognitive layer includes categories related to administration and system, language and script, and titles and communication, represented by words such as imperial examinations (in Chinese: 科举), oracle bone script (in Chinese: 甲骨文), and Lord (in Chinese: 老爷). These terms, which are largely associated with the historical and cultural system of the target language, are less familiar to learners because of their rarity in daily Life and Chinese historical documents, resulting in the lowest average cognitive score of 3.4671.

Relatedly, the hierarchical structure diverges from conventional patterns observed in widely adopted Chinese vocabulary outlines within the international education domain and classic textbooks in the Chinese language. Distinctive cultural lexemes, such as Tang poetry (in Chinese: 唐诗) and the Silk Road (in Chinese: 丝绸之路), represent the quintessence of Chinese culture. While not integral to the daily discourse of learners or contemporary Chinese youth, these terms enjoy heightened recognition among the student demographic. This phenomenon underscores the pivotal role of formal instructional methods and the deliberate inclusion of these terms in educational materials, shedding light on the profound influence of pedagogical resources on learners' cognitive grasp of cultural vocabulary. In contrast, learners of vocabulary that are heavily influenced by popular culture, even if they are more difficult and recognized as superlatives by the Standards, usually show a higher level of familiarity

in the test, e.g., TikTok (in Chinese: 抖音). This indicates a discrepancy between the difficulty in the Standards and the learners' actual perception of difficulty. The above observations suggest a sophisticated interplay between daily experiential nuances and the deliberate cultivation of formal educational content in shaping learners' comprehension of culturally significant terms (as shown in Fig. 2).

Interpretation of score clustering and variation range

Cluster analysis revealed that learners' cultural word recognition scores form effective clusters that are significantly correlated with their second language proficiency levels. In the initial tier, learners with an HSK level of 3 exhibit clustered cultural word recognition scores ranging from 2.8 to 3.6, indicating of relatively low precision and breadth in cultural word recognition. In the second tier, learners with an HSK level of 4 demonstrate clustered scores between 3.3 and 4, indicating improved breadth compared with Level 1 learners, albeit with continued modest accuracy. With an HSK level of 5, learners at Level 3 exhibit cultural word scores clustered between 3.7 and 4.5, indicating significantly enhanced accuracy and breadth of cultural word recognition compared with learners at Levels 1 and 2. In the fourth tier, learners with an HSK level of 6 or higher present cultural word scores concentrated in the range of 4.6 to 5, indicating substantially improved accuracy and breadth in cultural word recognition compared with learners in the preceding three tiers (as shown in Fig. 3).

The cultural connotations of words and their dissemination

The cultural significance and dissemination of lexical items can be illustrated through the examples of panda (熊猫), phoenix (凤凰), and qilin (麒麟). The phoenix and qilin, deeply rooted in Chinese mythology, possess rich symbolic meanings. The phoenix is

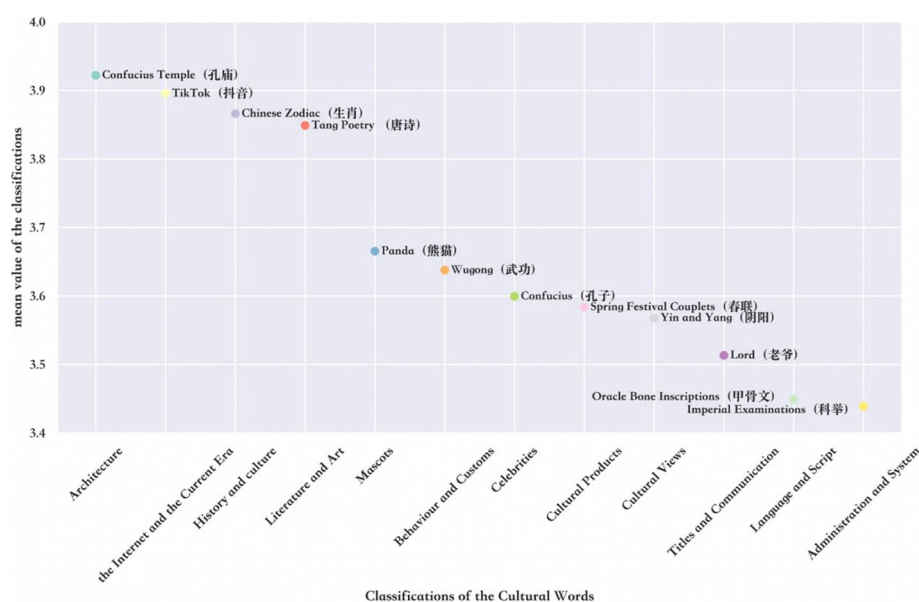


Fig. 2 Classification of the culture words

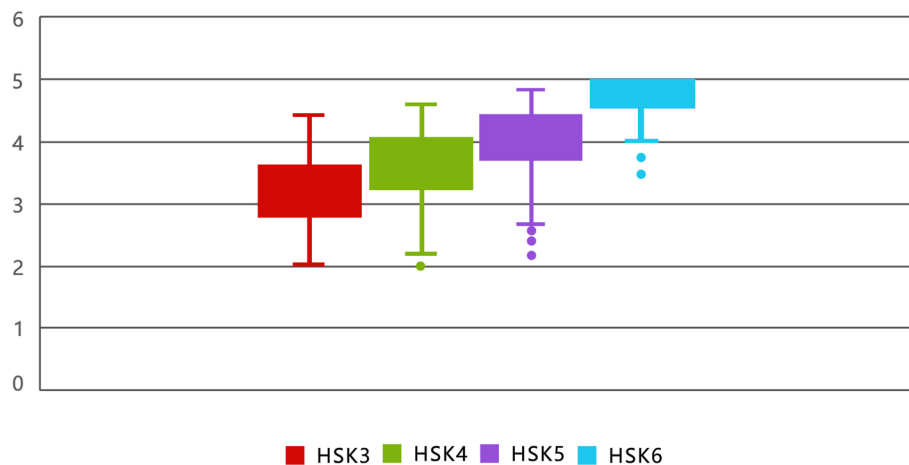


Fig. 3 Cultural word recognition scores of learners with different HSK levels

traditionally associated with grace, virtue, and prosperity, embodying harmony and often symbolizing the Empress. The qilin, likewise, represents auspiciousness and longevity, frequently linked to sages and virtuous rulers, and is celebrated for its gentle and noble nature. Both mythical creatures occupy central positions in Chinese cultural narratives and embody values that have been transmitted through generations.

In contrast, the panda, while lacking the same depth of historical or mythological significance, has gained immense contemporary cultural value. Its unique physical appearance, endangered status, and role as a symbol of wildlife conservation have elevated it to a globally recognized emblem of China. Among international learners, the panda often garners greater familiarity and emotional resonance than the phoenix or qilin, demonstrating how modern cultural symbols can transcend traditional narratives in terms of global dissemination and cross-cultural impact.

Cultural knowledge and cultural perception

The scores of certain cultural words may highlight disparities in learners' acquisition of Chinese cultural concepts and cultural perceptions. For example, terms highly significant to native Chinese speakers, such as "descendants of Yan and Huang" (a culturally loaded term for self-identification), exhibit relatively low cognitive recognition among second language learners. This suggests that while knowledge-based cultural concepts and words are readily acquired and comprehended, others tied to specific communicative contexts with nuanced meanings pose greater challenges, resulting in lower levels of cognitive recognition.

Effectiveness of Chinese character complexity

The statistical analysis results indicate that cultural words comprising Chinese characters with complex stroke structures tend to have relatively lower recognition scores, regardless of learners' proficiency levels. This underscores a notable correlation

between the complexity of Chinese characters and word recognition. Examples of such words include “couplet” (in Chinese 楹联), “qilin” (in Chinese 麒麟), “seal script” (in Chinese 篆书), and “taboo” (in Chinese 避讳).

The effectiveness and function of distraction words

When evaluating the depth and breadth of participants’ cultural word knowledge, researchers can integrate a cultural vocabulary scale with a list of distractor words. First, the cultural vocabulary scale serves as the fundamental tool for assessing participants’ vocabulary knowledge within specific cultural domains, enabling researchers to gauge their comprehension across various cultural vocabulary categories. Concurrently, distractor words encompass common and general vocabulary items categorized into elementary, intermediate, and advanced levels. By comparing participants’ self-reported Chinese language proficiency levels with their performance on distractor words, researchers can gauge the seriousness with which participants approached the questions. For example, if a participant with a higher Chinese language proficiency level assigns a low score to elementary-level distractor words, it may warrant excluding such questionnaire data. This approach optimizes test data quality.

The impact of cultural words on teaching and learning

Cultural vocabulary reflects a community’s values, traditions, and social norms, making it essential in language education. Its integration into curriculum design supports a gradual progression from basic communication to deeper cultural understanding. In International Chinese Education, for example, words like “孝顺” xiàoshùn (filial piety) and “面子” miànzi (face) convey key cultural concepts and should be introduced at appropriate learning stages.

This study reveals that the recognition of cultural words cannot be attributed solely to the graphic complexity of Chinese characters but is more fundamentally shaped by exposure frequency and contextual familiarity. Pedagogically, cultural vocabulary requires contextual explanation and the use of diverse resources such as stories, festivals, and media. It also involves pragmatics, including politeness and social conventions, demanding that teachers be culturally aware and pedagogically flexible.

For learners, cultural vocabulary enhances language skills, promotes intercultural awareness, and increases engagement. It supports not only linguistic development but also critical thinking and cultural comparison. Therefore, effective selection and instruction of cultural vocabulary are crucial for improving both language proficiency and intercultural competence.

Limitation

The cultural word scale developed in this study excludes the vocabulary specified in the Chinese International Education Proficiency Level Standards. However, it is important to recognize that the Standards also include many words with significant cultural richness. Future research should focus on further empirical studies concerning the selection of cultural words. Specifically, manual scoring is conducted on the cultural lexicon used in this study and the 11,092 words in the Standards to quantify the cultural meaning

richness of each word. This effort aims to provide a scientific basis for expanding and updating the cultural lexicon and the cultural word scale.

Moreover, while the Chinese Cultural Word Scale, based on the Likert scale, is intuitive, standardized, and adaptable, it has limitations that should not be overlooked. The scale might not fully capture the complexity of Chinese cultural words, and variations in cultural backgrounds could introduce biases during its application, potentially affecting data accuracy. Additionally, the presence of a middle option on the scale could lead to neutral responses, which might impact the precision of the results. Recognizing and addressing these potential shortcomings is crucial for improving the effectiveness of the scale's application.

Furthermore, the participant pool in this study primarily comprises individuals from the Chinese character culture circle, with limited representation from regions such as Europe, the USA, and South America. This relative homogeneity may limit the generalizability of the findings to learners from diverse cultural backgrounds. Therefore, future research should aim to include more participants from non-Chinese character cultural circles to increase the scale's validity and applicability across broader demographics. Additionally, future studies should explore the longitudinal effects of cultural vocabulary instruction on learners' proficiency and cultural awareness over time.

Conclusion and outlook

This study significantly contributes to the field of international Chinese education by establishing a cultural vocabulary lexicon grounded in robust multisource statistical data. Building upon this foundation, a Chinese cultural vocabulary scale has been successfully developed and validated to assess the proficiency of international Chinese language learners in cultural vocabulary. The empirical research conducted demonstrates the scale's exceptional reliability, structural validity, and empirical validity, effectively capturing the breadth and depth of learners' cognitive understanding of Chinese cultural vocabulary.

The findings of this research highlight the significant impacts of Chinese language proficiency, study duration, and learners' cultural backgrounds on cultural vocabulary knowledge. Specifically, Chinese language proficiency emerges as the most crucial factor in predicting self-assessment scores of cultural vocabulary, followed by study duration and cultural background. Conversely, gender does not have a notable effect on cultural vocabulary knowledge.

Furthermore, the study reveals variations in learners' knowledge of cultural vocabulary on the basis of frequency and temporal relevance. Learners exhibit greater knowledge of high-frequency and modern culture-related vocabulary, whereas their grasp of vocabulary with profound historical and cultural connotations is relatively lower. This discovery underscores the importance of prioritizing the teaching of cultural vocabulary in Chinese language education, particularly in developing learners' understanding and application abilities related to traditional cultural vocabulary.

In applying the cultural lexicon and vocabulary scale, the ongoing efforts to establish the International Chinese Education Graded Reader Standards are still nascent, and creating a cultural vocabulary list could positively influence this initiative. Moreover, developing a cultural vocabulary list can further refine the vocabulary outline within the

“Chinese Proficiency Level Standards for International Chinese Education.” By evaluating the cultural vocabulary proficiency of Chinese learners globally, valuable data and feedback can be garnered to enhance future iterations of the vocabulary outline in the “Standards,” aligning it more closely with learner requirements and thereby enhancing the quality and efficacy of international Chinese education.

Moreover, as previously noted, *The Framework of Reference for Chinese Culture and Society in International Chinese Language Education* (2022) serves as the first cultural teaching reference framework in the field of international Chinese education. However, this framework has been primarily constructed on the basis of expert knowledge. In the future, if researchers in international Chinese education integrate the cultural vocabulary scale as an empirical tool, it holds considerable promise for further optimizing this qualitatively grounded framework through empirical research. Such empirical data are crucial for refining the framework’s structure, content, and implementation, thereby advancing the high-quality development of cultural instruction in the field of international Chinese education.

Additionally, “The Framework of Reference for Chinese Culture and Society in International Chinese Language Education” (2022) is the first cultural teaching reference framework in the field of international Chinese education. Enumerating cultural points and describing cultural elements establishes the content and hierarchical system of cultural instruction. Although the “Reference Framework” fundamentally belongs to qualitative research, applying the cultural vocabulary scale, which measures the depth and breadth of cultural vocabulary for students worldwide, can provide empirical research support for its optimization. This, in turn, promotes the high-quality development of cultural instruction in the field of international Chinese education.

Overall, this study provides a valuable tool for the advancing of international Chinese education, contributing to the measurement of learners’ cultural vocabulary proficiency, fostering cross-cultural understanding, and establishing a solid foundation for future teaching and research endeavors. Through empirical validation, the scale has demonstrated its effectiveness in assessing the proficiency of Chinese second language learners in understanding cultural vocabulary. The findings underscore the importance of Chinese language proficiency, learning duration, and learners’ national cultural background in shaping their knowledge of cultural vocabulary. Moving forward, the scale can serve as a valuable tool for educators and researchers to evaluate learners’ cultural vocabulary skills. It is also recommended that future researchers revisit and expand upon this line of research to explore additional meanings and potential benefits of the findings, thereby strengthening its relevance and contributions to education. The large-scale cultural vocabulary measurement results are expected to inform curriculum design, teaching assessments, and research endeavors, thereby enhancing learners’ vocabulary skills and cross-cultural understanding in the field of international Chinese language education.

To further enhance the vocabulary scale, the following aspects may be considered:

Dynamically updating the international Chinese education cultural vocabulary corpus by monitoring internet news corpora and integrating new words and phrases reflecting traditional or contemporary Chinese cultural concepts.

Other measurement forms, such as word selection, filling in blanks, and word definitions, are integrated to enrich the measurement mode of learners' depth of knowledge of cultural vocabularies.

The cultural vocabulary scale was used to measure cultural attitudes and identify global learners of Chinese as a second language from different cultural spheres. This approach can aid in understanding their cognitive interest in and knowledge of Chinese culture. The statistical results derived from this analysis can be utilized for second language teaching and cultural research purposes.

Abbreviation

CS/FL Chinese as a second/foreign language learner

Acknowledgements

We gratefully acknowledge Professor Haiwei Zhang from Peking University for his valuable support and guidance in the data analysis of this study.

Authors' contributions

Lijiao Yang led the overall conceptualization, initial manuscript drafting, and project funding support. Yifan Lin was responsible for enhancing the introduction and related research sections, supplementing the literature review during the early stages, and participating in the final revision of the manuscript. Panpan Zhou provided key support in the final-stage editing and refinement. Danyang Qiu contributed to the early-stage development of the project's corpus. Luyao Chen contributed to the comprehensive revision of the manuscript, as well as data analysis and interpretation. Lijiao Yang and Yifan Lin contributed the same to this work as co-first-authors. Lijiao Yang and Luyao Chen share the co-corresponding authorship.

Funding

This research was supported by the General Program of the National Social Science Fund of China (Grant No. 23BY198).

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

Received: 27 August 2024 Accepted: 29 August 2025

Published online: 30 October 2025

References

- Altakhaine, A. R. M., & Zibin, A. S. (2014). Perception of culturally loaded words by Arab EFL learners. *International Journal of Linguistics*, 6(3), 1–22. <https://doi.org/10.5296/ijl.v6i3.4922>
- Baker, M. (2018). *In other words: A coursebook on translation* (3rd ed.). London: Routledge.
- Benattabou, D. (2020). On the teaching and learning of culture-loaded words: The case of Moroccan EFL learners. *International Journal of Linguistics and Translation Studies*, 1(3), 1–19. <https://doi.org/10.36892/ijlts.v1i3.62>
- Byon, A. S. (2007). The use of culture portfolio project in a Korean culture classroom: Evaluating stereotypes and enhancing cross-cultural awareness. *Language, Culture and Curriculum*, 20(1), 1–19. <https://doi.org/10.2167/lcc323.0>
- Carrell, P. L. (1987). Content and formal schemata in ESL reading. *TESOL Quarterly*, 21(3), 461–481. <https://doi.org/10.2307/3586498>
- Center for Language Education and Cooperation, Ministry of Education. (2021). *Guoji zhongwen jiaoyu zhongwen shuiping dengji biaozhun* [Chinese proficiency grading standards for international Chinese language education]. Beijing: Beijing Yuyan Daxue Chubanshe.
- Chang, J. Y. (1995). *Hanyu cihui yu wenhua* [Chinese vocabulary and culture]. Beijing: Beijing Daxue Chubanshe.
- Citron, J. L. (1995). Can cross-cultural understanding aid second language acquisition? Toward a theory of ethno-lingual relativity. *Hispania*, 78(1), 105–113. <https://doi.org/10.2307/345230>
- Crossley, S. A., Salsbury, T., McNamara, D. S., & Jarvis, S. (2011). What is lexical proficiency? Some answers from computational models of speech data. *Tesol Quarterly*, 45(1), 182–193. <https://doi.org/10.5054/tq.2010.244019>
- Dagut, M. B. (1978). Hebrew-English translation: A linguistic analysis of some semantic problems. University of Haifa.
- Dang, Z. S. (2015). Minsu wenhua ci de fanyi wenti tanwei—Cong *Hong Lou Meng* Yingwenban zhong “yasuiqian” deng ci de fanyi tanqi [On translation of Chinese folk cultural phrases: Taking Hawkes and Yang Xianyi's translation of *Ya Sui Qian* in *Hong Lou Meng* as an example]. *Waiyu Jiaoyu* [Foreign Language Education], 36(1), 93–97. <https://doi.org/10.16362/j.cnki.cn61-1023/h.2015.01.020>

- Hao, M. L., & Fan, H. Q. (2008). Bujian tezheng yu jiegou leixing dui liuxuesheng hanzi shuxie de yingxiang [The influence of component features and structural types on international students' Chinese character writing]. *Yuyan Jiaoxue Yu Yanjiu [Language Teaching and Linguistic Studies]*, 5, 24–31.
- Hartmann, R. R. K., & James, G. (2002). *Dictionary of lexicography*. Routledge.
- Jiang, X., & Liu, Y. (2004). Pinyin wenzi beijing de waiguo xuesheng hanzi shuxie cuowu yanjiu [A study of character writing errors by foreign learners using alphabetic writing]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]*, 1, 60–70.
- Koda, K. (1994). Second language reading research: Problems and possibilities. *Applied Psycholinguistics*, 15(1), 1–28. <https://doi.org/10.1017/S0142716400006950>
- Koda, K. (2005). Insights into second language reading: A cross-linguistic approach. *Cambridge University Press*. <https://doi.org/10.1017/CBO9781139524841>
- Lado, R. (1972). *Teaching English as a second language: Patterns of difficulty in vocabulary*. New York: McGraw-Hill.
- Lantolf, J. P., & Pavlenko, A. (1995). Sociocultural theory and second language acquisition. *Annual Review of Applied Linguistics*, 15, 108–124. <https://doi.org/10.1017/S0267190500002646>
- Laufer, B. (1998). The development of passive and active vocabulary in a second language: Same or different? *Applied Linguistics*, 19(2), 255–271. <https://doi.org/10.1093/applin/19.2.255>
- Laufer, B., & Ravenhorst-Kalovski, G. C. (2010). Lexical threshold revisited: Lexical text coverage, learners' vocabulary size and reading comprehension. *Reading in a Foreign Language*, 22(1), 15–30.
- Li, D. N. (2000). Hanguo liuxuesheng "wenhua ci" xuexi tedian tanxi—Jian lun dui Hanguo liuxuesheng de Hanyu cihui jiaoxue [An analysis of the learning characteristics of "cultural words" among Korean international students: Also on Chinese vocabulary teaching for Korean international students]. *Nanjing Daxue Xuebao (Zhhexue, Renwen Kexue, Shehui Kexue Ban) [Journal of Nanjing University (Philosophy, Humanities and Social Sciences)]*, (5), 142–147.
- Li, X. J. (2021). *Zhonghua chuantong wenhua jianming cidian* [Concise dictionary of traditional Chinese culture]. Beijing: Zhongguo Dabaike Quanshu Chubanshe.
- Li, X. J. (Ed.). (2021). *Zhonghua chuantong wenhua jianming cidian* [A concise dictionary of traditional Chinese culture]. Beijing: Zhongguo Dabaike Quanshu Chubanshe.
- Linacre, J. M. (2025). *Winsteps: Rasch measurement computer program* (5.10.2) [Computer software]. Winsteps.com. <https://www.winsteps.com>
- Mackey, A. (2004). The social turn in second language acquisition. *Studies in Second Language Acquisition*, 26(4), 639–641. <https://doi.org/10.1017/S0272263104350047>
- Meng, Z. M. (1992). Jiaoji wenhua yu duiwai Hanyu jiaoxue [Communicative culture and teaching Chinese as a foreign language]. *Yuyan Jiaoxue Yu Yanjiu [Language Teaching and Linguistic Studies]*, 1, 95–110.
- Ministry of Education Center for Language Education and Cooperation. (2022). *Guoji hanyu jiaoxue zhong de zhongguo wenhua yu shehui zhinan* [The framework of reference for Chinese culture and society in international Chinese language education]. Beijing: Sinolingua.
- Nation, I. S. P. (2001). Learning vocabulary in another language. *Cambridge University Press*. <https://doi.org/10.1017/CBO9781139524759>
- Nation, I. S. P. (2005). Teaching and learning vocabulary. In E. Hinkel (Ed.), *Handbook of research on second language teaching and learning* (pp. 581–596). Lawrence Erlbaum.
- Newmark, P. (1998). *More paragraphs on translation*. Multilingual Matters.
- Nida, E. A. (1964). *Toward a science of translating: With special reference to principles and procedures involved in Bible translating*. Leiden: Brill.
- Phoenix Net, & Dunhe Foundation. (2023). *Big data report on the popularity of Chinese traditional culture* (2022). <https://i.ifeng.com/c/8NKTy7JHLul>
- Qian, X. J. (2002). Cihuilang ceshi yanjiu chutan [A preliminary study on vocabulary size testing]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]*, 4, 54–62.
- Risager, K. (2006). *Language and culture: Global flows and local complexity*. Bristol: Multilingual Matters.
- Schmitt, N. (2008). Instructed second language vocabulary learning. *Language Teaching Research*, 12(3), 329–363. <https://doi.org/10.1177/1362168808089921>
- Schmitt, N., Schmitt, D., & Clapham, C. (2001). Developing and exploring the behaviour of two new versions of the Vocabulary Levels Test. *Language Testing*, 18(1), 55–88. <https://doi.org/10.1177/026553220101800103>
- Shen, H. (2009). Size and strength: Written vocabulary acquisition among advanced learners. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]*, 23(1), 74–85.
- Steffensen, M. S., Joag-Dev, C., & Anderson, R. C. (1979). A cross-cultural perspective on reading comprehension. *Reading Research Quarterly*, 15(1), 10–29. <https://doi.org/10.2307/747429>
- Tišma, D. M. J., & Kršmanović, A. R. (2018). The effect of cultural awareness and hemispheric preference on Serbian EFL learners' vocabulary proficiency. *Nasleđe : Journal of Language, Literature, Arts & Culture*, 14(39), 1–15.
- Tüfekçioğlu, B. (2021). Cultural vocabulary in teaching Turkish as a foreign language. *Eurasian Journal of Applied Linguistics*, 7(1), 338–358. <https://doi.org/10.32601/ejal.911449>
- Wang, D. C. (1998). Guosu yuyi he Hanyu Guosu Cidian [National popular semantics and the National Popular Dictionary of the Chinese Language]. In W. Y. Fu (Ed.), *Guosu yuyi yanjiu* [Research on national popular semantics] (pp. 123–135). Shanghai Foreign Language Education Press.
- Wright, B. D., & Stone, M. H. (2004). *Making measures*. Chicago: The Pharon Press.
- Wu, Z. W. (2022). Zhongguo wenhua jiaoxue yu Zhongwen jiaoxue de jiehe tuijin [Approaches to integrating Chinese culture teaching with Chinese language teaching]. *International Journal of Chinese Language Teaching*, 2(2), 8–15.
- Yang, L. J. (2019). *Guoji Hanyu Jiaocai zhong de huati sheji yu chuli* [Topic design and handling in international Chinese language textbooks]. Beijing: Zhongyang Bianyi Chubanshe.
- Yu, L. X. (2016). Mao Zedong shici Yingyi ben zhong wenhua fuheci fanyi de duibi yanjiu [A comparative study on the translation of culture-loaded words in English versions of Mao Zedong's poems]. *Waiguoyu Xuekan* [Foreign Language Research], (6), 106–109. <https://doi.org/10.16263/j.cnki.23-1071/h.2016.06.025>
- Zhang, H. J. (2003). Shilun Hanyu ciyu de wenhua fujiayi [On the cultural connotations of Chinese words]. *Hanyu Xuexi [Chinese Language Learning]*, 3, 45–48.

- Zhang, H. S. (2006). Waiguo xuesheng Hanyu cihui xuexi zhuangkuang jiliang yanjiu [A quantitative study on the vocabulary learning situation of foreign students in Chinese]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]*, 1, 70–76.
- Zhang, H. W., Zhang, X. Y., Zhang, T. J., & Wang, R. X. (2021). Liuxuesheng shizi liangbiao bianzhi yanjiu [The creation and validation of a Hanzi recognition size test for learners of Chinese as a second language]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]*, 35(1), 126–142. <https://doi.org/10.13724/j.cnki.ctiw.2021.01.013>
- Zhang, S. H. (2012). Hanzi quan yu fei Hanzi quan liuxuesheng wenhua ci xide bijiao yanjiu [A comparative study on the acquisition of cultural words by students from Chinese character and non-Chinese character backgrounds]. *Yunnan Shifan Daxue Xuebao (Duiwai Hanyu Jiaoxue yu Yanjiu Ban) [Journal of Yunnan Normal University (Teaching and Research on Chinese as a Foreign Language Edition)]*, 10(6), 63–69. <https://doi.org/10.16802/j.cnki.ynsddw.2012.06.009>
- Zhao, G. (2003). Chuji jieduan Oumei liuxuesheng shizi liang yu zi de gouci shu [The literacy levels and morpheme productivity of European and American students at the elementary stage]. *Yuyan Wenzhi Yingyong [Applied Linguistics]*, (3), 106–112. <https://doi.org/10.16499/j.cnki.1003-5397.2003.03.017>
- Zhao, M. (2009). Guanyu Hanyu wenhua ci dingyi yanjiu de huigu [A review of studies on the definition of Chinese cultural words]. *Xiandai Hanyu [modern Chinese]*, 4, 7–9.
- Zhao, M. (2010). Duiwai Hanyu zhongji jiaocai zhong wenhua ci jieshou du de lianghua yanjiu [A quantitative study of cultural word acceptance in Chinese as a foreign language intermediate textbooks]. *Huaqiao Jiaoyu [overseas Chinese Education]*, 1, 9–15.
- Zu, X. M. (2023). Wenhua jiaoxue de xin linian he xin silu—《Guoji Zhongwen Jiaoyu Yong Zhongguo Wenhua Guoqing Jiaoxue Cankao Kuangjia》 jiedu [New concepts and approaches in cultural teaching: An interpretation of the Reference Framework for Chinese Culture and Society in International Chinese Language Education]. *Yuyan Jiaoxue Yu Yanjiu [Language Teaching and Linguistic Studies]*, 3, 26–35.
- Zu, Y. Q. (Ed.). (2022). *Guoji zhongwen jiaoyu zhonghua wenhua he shehui cankao kuangjia* [Framework of reference for Chinese culture and society in international Chinese language education]. Beijing: Yuhe Chubanshe.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.